Veterinary Package Insert

TP-SQ 250 WSP SULFAQUINOXALINE 250MG/G POWDER (25% Sulfaquinoxaline)

PRODUCT DESCRIPTION

Yellow colour powder, which contains 25% of sulfaquinoxaline) in 500gm or 1kg sachet.

PHARMACODYNAMICS & PHARMACOKINETICS

Sulfaquinoxaline belongs to a group of antibiotics called the sulphonamides. It is bacteriostatic. Sulfonamides interfere with the biosynthesis of folic acid in bacterial cells; they compete with para-aminobenzoic acid (PABA) for incorporation in the folic acid molecule. By replacing the PABA molecule and preventing the folic acid formation required for DNA synthesis, the sulfonamides prevent multiplication of the bacterial cell. Only organisms that synthesize their own folic acid are susceptible; mammalian cells use preformed folic acid and, therefore, are not susceptible. Cells that produce excess PABA or environments with PABA, such as necrotic tissues, allow for resistance by competition with the sulfonamide.

Absorption

Most sulfonamides are well absorbed orally with the exception of the enteric sulfonamides, such as sulfaquinoxaline, which are minimally absorbed. Delays in absorption may occur in adult ruminants or when sulfonamides are administered with food to monogastric animals.

Distribution

Sulfonamides are widely distributed throughout the body. They cross the placenta, and a few penetrate into the cerebrospinal fluid. Sulfonamides may be distributed into milk; however, they vary greatly in their ability to do so. The process depends on several factors, including protein binding and pKa values.

Protein Binding

Binding can vary depending on serum concentration and other factors.

Biotransformation

Sulfonamides are primarily metabolized in the liver but metabolism also occurs in other tissues. Biotransformation occurs mainly by acetylation, glucuronide conjugation, and aromatic hydroxylation in many species. The types of metabolites formed and the amount of each varies depending on the specific sulfonamide administered; the species, age, diet, and environment of the animal; the presence of disease; and, with the exception of pigs and ruminants, even the sex of the animal. Dogs are considered to be unable to acetylate sulfonamides to any significant degree.

N_4-acetyl metabolites have no antimicrobial activity and hydroxymetabolites have 2.5 to 39.5% of the activity of the parent compound. Metabolites may compete with the parent drug for involvement in folic acid synthesis but have little detrimental effect on the bacterial cell, and so could lower the activity of the remaining parent drug.

Duration of Action

The sulfonamides have been loosely categorized according to their plasma concentration versus time profiles:

Intermediate- to long-acting—Sulfadimethoxine, sulfamethazine. Enteric (minimally absorbed)—Sulfaquinoxaline.

Note: Duration of action may be estimated by the length of time target serum concentrations are maintained. Target concentrations are generally based on minimum inhibitory concentrations for each organism. Many sources use 50 mcg sulfonamide per mL (5 mg per decaliter) of blood as the minimum effective concentration for sulfonamides in animals.

Elimination

Renal excretion is the primary route of elimination for most nonenteric sulfonamides and it occurs by glomerular filtration of parent drug, tubular excretion of unchanged drug and metabolites, and passive reabsorption of nonionized drug. Alkalization of the urine increases the fraction of the dose that is eliminated in the urine. In general, the metabolites of the parent drug are more quickly eliminated by the kidney than the original sulfonamide is, but the proportions of metabolites formed can vary, depending on many factors.

INDICATION

In chickens and turkeys:

For the treatment of coccidia in the early stage of infection.

RECOMMENDED DOSAGE AND ADMINISTRATION

To be administered orally.

For use in drinking water.

Chickens & turkeys:

60 mg per kg body weight Sulfaquinoxalin per day according to 0.24 g per kg of body weight of product per day.

3 days of care, 2 days off, 2 days treatment.

In the dosage above which is a mixing ratio of product in the drinking water for the animals to be treated according to the following formula to calculate:

$$\frac{\text{... mg of product per kg bw / day} \quad \text{Mean body weight (kg) of animals to be treated}}{\text{Mean daily water consumption (l) / animal}} = \text{… mg of product per liter of drinking water}$$

CONTRAINDICATIONS
Severe hepatic and renal impairment .

Diseases that or with greatly reduced fluid intake, accompanied by severe fluid losses.

Damage to the hematopoietic system.

Hypersensitivity to sulphonamides.

Resistance to sulphonamides.

Do not use in chickens and turkeys which eggs are produced for human consumption.

WARNINGS & PRECAUTIONS.
Special precautions for use in animals
Ensure sufficient water intake during treatment. Due to the low compatibility the product, it should be used only if clearly needed. The use of product should be based on susceptibility testing.

Special precautions for user
To avoid sensitization or contact dermatitis, direct skin contact and inhalation during processing or use during treatment are to be avoided. Use dust mask and gloves.

INTERACTIONS WITH OTHER MEDICATIONS
Drug interactions relating specifically to the use of sulfonamides in animals are rarely reported in veterinary literature.

Mixing with other drugs should be avoided because of possible incompatibilities.

PREGNANCY AND LACTATION
For sulfonamides safe use has not been established during pregnancy. They should be used only if the benefits of treatment clearly outweigh the risks.

Do not use in chickens and turkeys, which eggs are produced for human consumption.

ADVERSE EFFECTS
Liver damage, allergic reaction and blood disorders may occur after using the product.

The active ingredient of product, Sulfaquinoxaline may be relatively poor compatibility in chickens. Even under normal intermittent treatment, it may develop hemorrhagic syndrome with bleeding in subcutaneous and muscles, disturbed general condition, thrombocytopenia and even death. The addition of vitamin K does not prevent the formation of this syndrome. Several chicken flock deaths occur under treatment with Sulfaquinoxaline which probably caused by an allergy-related panmyelopathy.

When administered via the drinking water, the change in flavor occurs and hence decrease in water intake. The weight reduction associated therewith can be avoided by intermittent administration of the product.

In laying hens, breeding egg production can lead to a drop in egg production after treatment with the product. Furthermore, egg shell and hatchability may be affected.

If the side effects mentioned above discontinue the product immediately.

OVERDOSE & TREATMENT
In overdose ataxic movements, muscle twitching, convulsions and comatose states may occur. Discontinue the product immediately.

The remaining product in the stomach should be removed by saline laxatives.

The neurotropic effects should be treated symptomatically by administering centrally sedative substances (for example, barbiturates).

In additional administration of vitamin K or folic acid would increase the renal excretion of sulfonamide. Excretion is indicated by alkalizing agents (for example sodium bicarbonate).

WITHDRAWAL PERIOD
Meat and offal: 14 days.

Do not use in chickens and turkeys which eggs are produced for human consumption.

STORAGE CONDITION
Store below 30°C. Protect from direct sunlight.

SHELF LIFE
2 years.
After 1st opening, reconstitution or dilution, use within 24 hours.

MAXIMUM RESIDUAL LIMIT (MRL)

Substance / Drug Definition of residue in which MRL was set	Food	Maximum Residue Limits (MRLs) in food µg/kg
Sulfaquinoxaline	Edible offal, muscle (poultry)	100

PACKING

500g

1kg

THYE PHARMA SDN. BHD.
(manufacturer & product registration holder)
No.2 Lorong Industri Ringan Permatang Tinggi 8,
Kawasan Industri Ringan Permatang Tinggi,
14000 Bukit Mertajam,
Pulau Pinang.

Revised August 2016